

LDV_DSL

Diesel Light Duty Vehicle



Diesel Car

Inputs / Outputs

dir	comm	unit	group
Input	DSL	PJ	i
Output	PLDVHWY	MPKm	o
Output	PLDVCTY	MPKm	o

Efficiency (ceff)

	comm	use2cact	cact2cout	share.up
2	PLDVHWY	686.083	2.03	0.4
3	PLDVCTY	506.547	2.03	0.8

Key Parameters

cap2act	12.04
olife	16 years
stock	7216.70 1000 Vehicles
cap unit	1000 Vehicles
act unit	million km, city
cost unit	USD/(Vh*km)

Fuel Efficiency

Mode	Fuel	L/100 km	km/kWh	MJ/km	mpg (US)
Highway	DSL	3.95	2.47	1.46	59.55
City	DSL	5.35	1.82	1.97	43.97

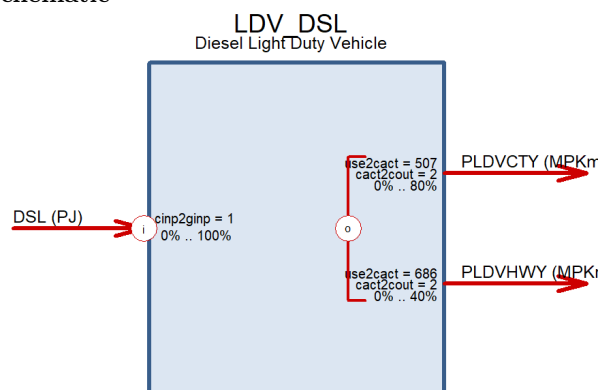
Vehicle Parameters

Fuels	DSL	Primary fuel	DSL
Efficiency (highway)	3.95 L/100km	Efficiency (city)	5.35 L/100km
Annual distance	12,043 km/yr	Capacity unit	1000 vehicles
Load/vehicle (hwy)	2.03 passengers	Load/vehicle (city)	2.03 passengers
Max share (highway)	40%	Max share (city)	80%
Vehicle cost	11,171 USD	Annual fixed cost	25 USD/yr
Lifetime	16 yr		

Capacity bounds

region	year	stock	cap.lo	cap.up	cap.fx	ncap.lo	ncap.up	ncap.fx	ret.lo	ret.up	ret.fx
IND	2024	7216.7	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2025	6765.7	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2030	4510.4	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2035	2255.2	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2040	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2045	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2050	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA

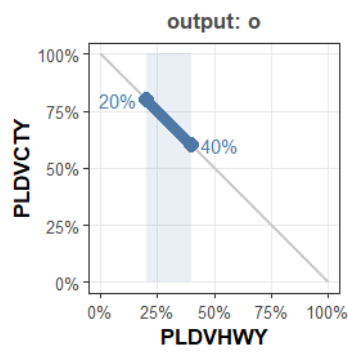
Schematic



Costs [USD/(Vh*km)]

Fixed O&M	0.02
Investment cost	11.17

Share Mix Constraints



Levelized Cost 0.17 [USD/(Vh*km)], 0.08–0.09 [USD/(Passenger*km)]

